

Challenge #1

1. Define the problem

As mentioned in the module, the problem has been defined: How can we identify countries with the biggest health tech market sizes?

2. Get the Data

The screenshot shows the World Bank DataBank interface. On the left, under 'Variables', two series are selected: 'Current health expenditure (% of GDP)' and 'Current health expenditure per capita (current US\$)'. On the right, the 'Preview' section shows data for Afghanistan in 2018. The data is as follows:

	2018
Current health expenditure (% of GDP)	14.21
Current health expenditure per capita (current US\$)	71.33

Source: World Development Indicators. Click on a metadata icon for original source information to be used for citation.

Then, I downloaded the data according to the desired data type; here, I downloaded the data in Excel format.

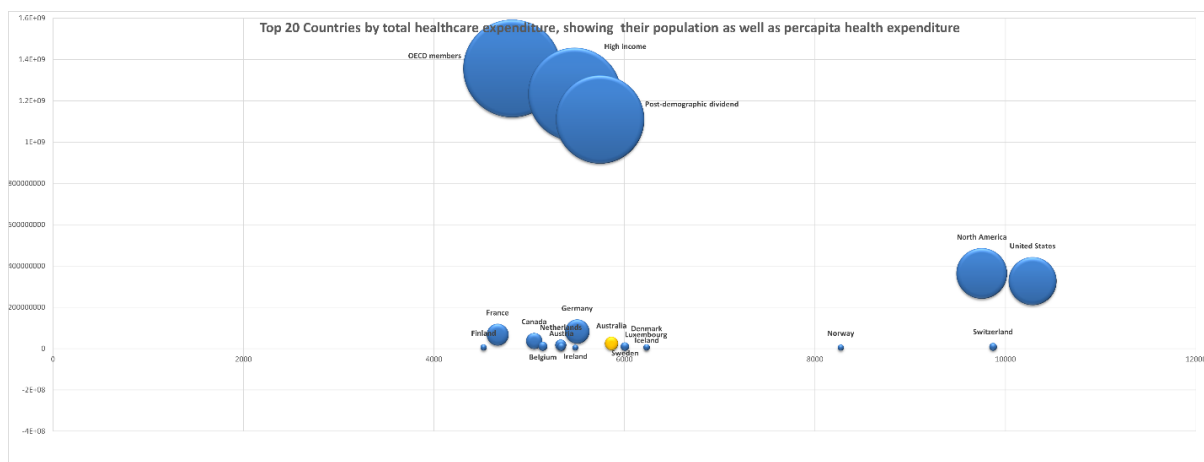
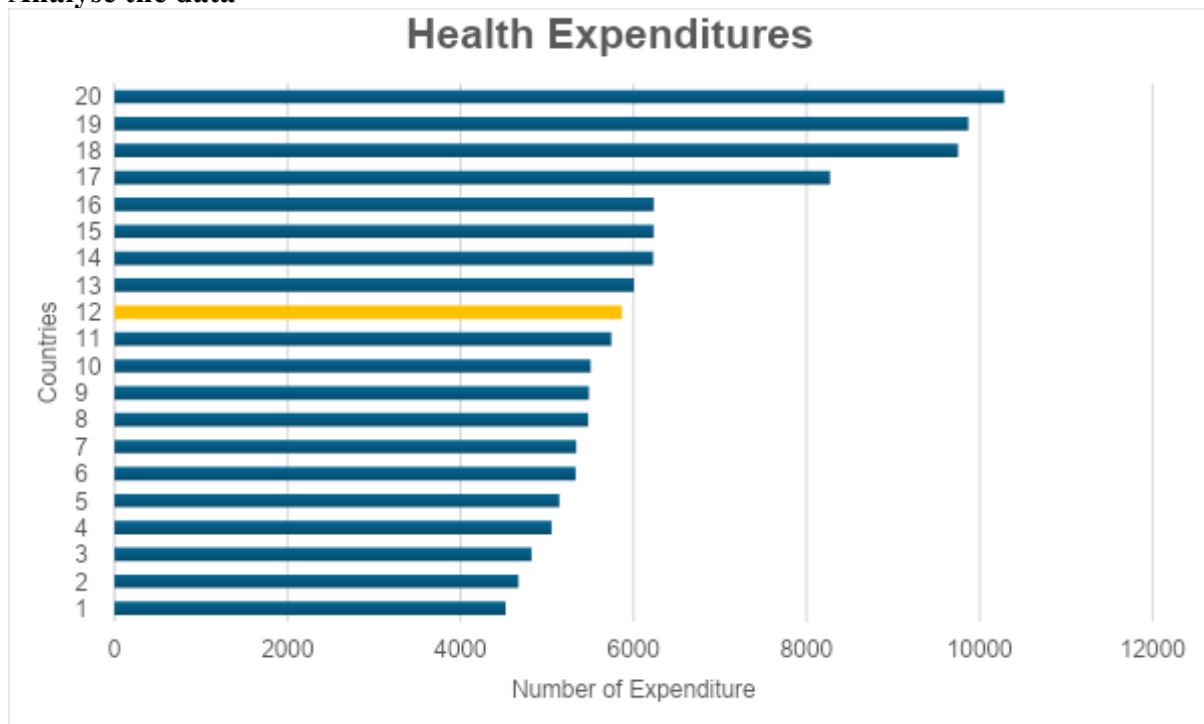
3. Clean and manipulate the data

There is missing data in the 2018 total expenditure column, we can drop it by applying a filter on all columns then unchecking the missing data.

The screenshot shows an Excel spreadsheet with a filter applied to the 2018 [YR2018] column. The filter menu is open, showing a list of values to be filtered. The values are: 6227, 6233, 6234, 7848.356445, 8272, 9212.432617, and 10280.52637. The filter is set to '2018 [YR2018]' and the 'Clear Filter From "2018 [YR2018]"' option is selected. The spreadsheet shows columns A through G, with rows 1 through 24. The 2018 [YR2018] column is highlighted in blue.

Apply this to all columns to get clean data for the analysis stage.

4. Analyse the data



5. Communicate the data

This dataset is suitable for the government, especially the health ministry, to improve the quality of health services. With this dataset, we can do the most important initial stage, including finding information on countries' health expenditures worldwide.

We assume that a country's health expenditures are usually correlated with an increase in the quality of health services and broader community access.

We should analyse health care expenditure before further analysing the quality of health services and facilities.

In Figure 1, the top three countries with the most significant healthcare expenditures are the United States, Switzerland, and North America, while Australia is in 9th position. It is also crucial to analyse each country's population.

In Figure 2, the amount of health expenditure is not fully proportional to a country's population; for example, Switzerland has a population smaller than North America and the United States but a more significant healthcare expenditure than those countries.

I recommend further analysing the quality of health services and facilities in Switzerland, North America, and the United States.